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Problem statement: Implement all the functions of a dictionary (ADT) using hashing and handle collisions using chaining with / without replacement.Data: Set of (key, value) pairs, Keys are mapped to values, Keys must be comparable, Keys must be unique. Standard Operations: Insert(key, value), Find(key), Delete(key)

```
def simple_hash(key, size):
    return sum(ord(c) for c in key) % size

class Dictionary:
    def __init__(self, size, mode="with"):
        self.size = size
        self.mode = mode # "with" or "without"
        self.table = [[] for _ in range(size)]

    def insert(self, key, value):
        index = simple_hash(key, self.size)
        for i in range(len(self.table[index])):
            if self.table[index][i][0] == key:
                if self.mode == "with":
                    self.table[index][i] = (key, value)
                    print(f"Updated key '{key}' with new value.")
                else:
                    print(f"Key '{key}' already exists. Skipping insert
(without replacement).")
                    return
            self.table[index].append((key, value))
            print(f"Inserted key '{key}' successfully.")

    def find(self, key):
        index = simple_hash(key, self.size)
        for k, v in self.table[index]:
            if k == key:
                print(f"Found: {key} => {v}")
                return v
        print(f"Key '{key}' not found.")
        return None

    def delete(self, key):
        index = simple_hash(key, self.size)
        for i in range(len(self.table[index])):
            if self.table[index][i][0] == key:
                del self.table[index][i]
                print(f"Deleted key '{key}'.")
                return True
        print(f"Key '{key}' not found for deletion.")
        return False

    def display(self):
        print("\nCurrent state of the dictionary:")
        for i in range(self.size):
            print(f"Index {i}: {self.table[i]}")
```

```

        print()

def main():
    print("=== Dictionary ADT using Hashing with Chaining ===")
    size = int(input("Enter size of the hash table: "))

    print("\nChoose collision handling mode:")
    print("1. With replacement (updates existing keys)")
    print("2. Without replacement (ignores duplicate keys)")
    mode_choice = input("Enter 1 or 2: ")

    if mode_choice == "1":
        mode = "with"
    elif mode_choice == "2":
        mode = "without"
    else:
        print("Invalid choice. Defaulting to 'with replacement'.")
        mode = "with"

    d = Dictionary(size, mode)

    while True:
        print("\nOptions:")
        print("1. Insert (key, value)")
        print("2. Find (key)")
        print("3. Delete (key)")
        print("4. Display Dictionary")
        print("5. Exit")

        choice = input("Enter your choice (1-5): ")

        if choice == "1":
            key = input("Enter key: ")
            value = input("Enter value: ")
            d.insert(key, value)
        elif choice == "2":
            key = input("Enter key to find: ")
            d.find(key)
        elif choice == "3":
            key = input("Enter key to delete: ")
            d.delete(key)
        elif choice == "4":
            d.display()
        elif choice == "5":
            print("Exiting program.")
            break
        else:
            print("Invalid choice. Try again.")

if __name__ == "__main__":
    main()

```

Output:

```
S C:\Users\Manav> === Dictionary ADT using Hashing with Chaining ===
> Enter size of the hash table: 5
>
> Choose collision handling mode:
> 1. With replacement (updates existing keys)
> 2. Without replacement (ignores duplicate keys)
> Enter 1 or 2: 1
>
> Options:
> 1. Insert (key, value)
> 2. Find (key)
> 3. Delete (key)
> 4. Display Dictionary
> 5. Exit
> Enter your choice (1-5): 1
> Enter key: Alice
> Enter value: 12345
> Inserted key 'Alice' successfully.
>
> Options:
> 1. Insert (key, value)
> 2. Find (key)
> 3. Delete (key)
> 4. Display Dictionary
> 5. Exit
> Enter your choice (1-5): 1
> Enter key: Bob
> Enter value: 67890
> Inserted key 'Bob' successfully.
>
```